

End-to-End Data Analytics Platform

Microsoft Fabric | Public Data | Spain Indicators

Complete functional specification defining analytical objectives, source-to-target mappings, star schema design, derived measures, data quality rules, and Power BI report specification.

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6 years experience	Lakehouse Warehouse	Duration: 1-2 months
Microsoft Stack	PySpark Power BI	Version 1.2

1. Executive Summary

This document defines the functional specification for the End-to-End Data Analytics Platform. It translates the architecture design (Phase 2) into concrete business definitions: what each data element means, how it is transformed across layers, what business rules govern the derived measures, and what the final analytical output looks like in Power BI.

The functional specification serves as the **contract between business intent and technical implementation**. Every field, every transformation rule, and every validation criterion defined here will be implemented in Phase 5 (Development) exactly as specified. Where the architecture document (Phase 2) defined WHICH components handle each step, this document defines WHAT those components must produce.

The document covers: analytical objectives that justify the platform, complete source-to-target mappings for all five data sources across the Bronze-Silver-Gold medallion layers, the star schema specification with all dimension and fact table definitions, business rules for the four derived measures (YoY, MoM, YTD, QTD), data quality and operational rules including the logging framework, the Power BI report specification, and a traceability matrix linking requirements to deliverables.

Audience: *This document is written for a functional audience: analysts, product owners, and stakeholders who need to validate WHAT the system does. Technical implementation details (DDL statements, PySpark code, connection configurations) are delivered in Phase 4 (Technical Specification).*

2. Analytical Objectives

The platform exists to answer specific analytical questions about Spain's economic landscape across five domains: consumer prices, industrial production, energy demand, public construction investment, and tax revenue.

2.1 Cross-Domain Analysis

The unified star schema enables questions that span multiple domains, which is the primary differentiator of this platform versus querying each source independently:

- How does industrial production (IPI) in Catalonia correlate with energy demand evolution in the same region?
- When consumer prices (IPC) rise sharply, does public construction investment increase or decrease in the following quarters?
- Is there a seasonal pattern where tax revenue peaks align with or diverge from industrial production peaks?
- How do Madrid and Catalonia compare across all five economic indicators over the last decade?

2.2 Single-Domain Deep Analysis

Domain	Source	Key Analytical Questions
Consumer Prices	IPC (INE)	Which ECOICOP categories drive inflation? How does regional inflation differ from national?
Industrial Production	IPI (INE)	Which economic destinations show growth/decline? How does the industrial cycle compare across regions?
Energy Demand	REE API	What is the demand trend? Are there structural shifts in energy consumption patterns?
Construction	MITMA	How does public investment evolve? State vs. Territorial Entities allocation?
Tax Revenue	AEAT	How does tax collection evolve by concept? Which categories drive revenue growth?

2.3 Temporal Analysis

All five sources share a temporal dimension with monthly granularity, enabling four standardized rates of change: Year-over-Year (YoY), Month-over-Month (MoM), Year-to-Date cumulative (YTD), and Quarter-to-Date cumulative (QTD). These measures allow direct comparison of momentum across domains regardless of the underlying unit of measure.

3. Source-to-Target Mapping

This section defines the complete data journey for each of the five sources across the Bronze-Silver-Gold medallion layers. For each source: what fields arrive in Bronze (raw), what transformations produce Silver (cleaned), and how Silver maps to Gold (star schema).

Convention: All source fields in Bronze are stored as string (text) type, regardless of their original content. Type casting to numeric, date, or other types is performed in the Silver layer. This protects the pipeline from source format changes that could break ingestion.

■ ■ **Encoding:** INE CSV files use ISO-8859-15 encoding. During Bronze ingestion, files are read with explicit ISO-8859-15 encoding and stored as UTF-8. This ensures characters like ñ, á, ü render correctly. All Silver and Gold tables use UTF-8 exclusively.

Source 1: Consumer Price Index (IPC)

CSV

Instituto Nacional de Estadística (INE)

The IPC dataset arrives as a semicolon-delimited CSV with ISO-8859-15 encoding. It contains the Consumer Price Index disaggregated by Autonomous Community, ECOICOP classification group, and data type. Only the index value (Indice) is retained; pre-calculated variations from the source are discarded because the platform computes its own standardized measures (YoY, MoM, YTD, QTD) uniformly across all sources.

Bronze Layer — spi_bronze_ipc_raw

Raw CSV loaded after encoding resolution. All fields stored as string. Metadata columns added.

Column	Origin	Description	Storage
comunidades_y_ciudades_autonomas	Source CSV	Autonomous Community name as provided by INE	String
grupos_ecoicop	Source CSV	ECOICOP classification group for the price index	String
tipo_de_dato	Source CSV	Type of data: Indice, Variacion mensual, etc.	String
periodo	Source CSV	Period in INE format: YYYYMMNN (e.g., 2026M02)	String
total	Source CSV	Numeric value (stored as text in Bronze)	String
_ingestion_timestamp	System	UTC timestamp when the row was ingested	Timestamp
_source_file	System	Name of the source file ingested	String

Silver Layer — spi_silver_ipc_cleaned

Filtered, renamed, typed. Only tipo_de_dato = 'Indice'. Geographic filter: Nacional, Cataluña, Madrid.

Column	Transformation Rule	Description
region_name	Mapped from comunidades_y_ciudades_autonomas. Normalized to: Nacional, Cataluña, Madrid	Standardized geographic entity
indicator_category	Renamed from grupos_ecoicop	ECOICOP group
year	Extracted from periodo: first 4 chars. Cast to int	Year of the observation
month	Extracted from periodo: chars after 'M'. Cast to int	Month (1-12)
indicator_value	Renamed from total. Cast to decimal	Index value

Filtering rules: Rows where tipo_de_dato is not 'Indice' are discarded. Rows where the geographic entity does not match the three entities are discarded. Rows where total cannot be cast to a valid number are discarded and logged.

Gold Mapping — spi_fact_indicators

Each Silver row produces one IDX row in the fact table. The Gold notebook then calculates YoY, MoM, YTD, and QTD, producing up to five rows per Silver row. The mapping to the fact table works through dimension lookups:

How dimension lookups work: For each Silver row, the notebook searches the dimension tables to find the matching surrogate key. For example, if Silver has region_name = 'Cataluña', the notebook looks up dim_region where region_name = 'Cataluña' and retrieves region_key = 2. That key (2) is written as the foreign key in the fact table.

Fact Column	Dimension	Lookup Logic
indicator_key	dim_indicator	Find row where domain = 'Consumer Prices' AND indicator_category matches Silver value
region_key	dim_region	Find row where region_name matches Silver value
calendar_key	dim_calendar	Find row where year and month match Silver values
measure_key	dim_measure	measure_code = 'IDX' for original index; derived measures use their own codes
source_key	dim_source	source_code = 'IPC'
indicator_value	Silver	indicator_value from Silver (unchanged for IDX; calculated for derived measures)

CSV → SQL

Source 2: Industrial Production Index (IPI)

Instituto Nacional de Estadística (INE) — via Azure SQL Database

The IPI dataset is pre-loaded into an Azure SQL Database staging table to demonstrate SQL Server ingestion. The original format is identical to IPC (INE CSV). Only the index value (Indice) is retained.

Bronze Layer — spi_bronze_ipi_raw

Column	Origin	Description	Storage
comunidades_y_ciudades_autonomas	Azure SQL	Autonomous Community name	String
destino_economico_de_los_bienes	Azure SQL	Economic destination classification	String
tipo_de_dato	Azure SQL	Type of data: Indice, Variacion mensual, etc.	String
periodo	Azure SQL	Period in INE format: YYYYMMNN	String
total	Azure SQL	Numeric value (stored as text)	String
_ingestion_timestamp	System	UTC timestamp of ingestion	Timestamp
_source_table	System	Name of the Azure SQL source table	String

Silver Layer — spi_silver_ipi_cleaned

Column	Transformation Rule	Description
region_name	Mapped and normalized: Nacional, Cataluña, Madrid	Standardized geographic entity
indicator_category	Renamed from destino_economico_de_los_bienes	Economic destination
year	Extracted from periodo, cast to integer	Year
month	Extracted from periodo, cast to integer	Month (1-12)
indicator_value	Renamed from total, cast to decimal	Index value

Filtering and Gold mapping: Identical pattern to IPC. source_code = 'IPI'.

REST API

Source 3: Electricity Demand Evolution

Red Eléctrica de España (REE) — Redeia Group

The energy demand data is ingested via the REE public REST API. The API returns nested JSON following the JSONAPI standard. Only 'Demanda real' (actual demand) is retained, representing realized electricity demand in MWh. Three API calls per execution (one per geographic entity).

API Response Structure

The JSON response contains a top-level 'included' array. Each element represents an indicator with an 'attributes.values' array of time-series data points. The relevant path is: `included[] → attributes.title` (filter: 'Demanda real') `→ attributes.values[] → {datetime, value}`.

API Parameters per Geographic Entity

Entity	geo_limit	geo_ids	time_trunc
Nacional	peninsular	8741	month
Cataluña	ccaa	9	month
Madrid	ccaa	13	month

Pagination: The API has a maximum date range per request (~1 year). The notebook iterates year by year from 2014 to current year. Each year-entity combination = one API call.

Bronze Layer — spi_bronze_energy_raw

Column	Origin	Description	Storage
datetime	API: values[].datetime	ISO 8601 timestamp	String
value	API: values[].value	Demand value in MWh (as text)	String
indicator_title	API: attributes.title	Indicator name (e.g., 'Demanda real')	String
geo_name	Derived from geo_ids	Geographic entity name	String
_ingestion_timestamp	System	UTC timestamp of ingestion	Timestamp
_source_file	System	Identifier: 'ree_api_demand'	String

Silver Layer — spi_silver_energy_cleaned

Column	Transformation Rule	Description
region_name	Mapped from geo_name. Normalized	Standardized geographic entity
indicator_category	Hardcoded: 'Demanda real'	Electricity demand indicator
year	Extracted from datetime, cast to integer	Year
month	Extracted from datetime, cast to integer	Month (1-12)
indicator_value	Renamed from value, cast to decimal	Electricity demand in MWh

Filtering: Only indicator_title = 'Demanda real' retained. Duplicates removed, keeping latest. Null/zero values discarded. Gold mapping: source_code = 'ENERGY'.

XLS

Source 4: Construction Tenders

Ministerio de Transportes y Movilidad Sostenible (MITMA)

The construction tenders data arrives as four legacy XLS files (Excel 97-2003 binary format) with merged cells, multi-row headers, and embedded subtotals. This is the highest-complexity ingestion source in the project.

File Inventory and Consolidation

File	Contracting Agent	Geographic Scope
01401400.XLS	State & Social Security	Nacional + Cataluña
01401600.XLS	State & Social Security	Madrid
01402600.XLS	Territorial Entities	Nacional + Cataluña
01402800.XLS	Territorial Entities	Madrid

Consolidation tool: The Bronze ingestion notebook (spi_nb_bronze_construction) uses the Python library xlrd to read each XLS file, parse the multi-row headers and merged cells programmatically, derive the contracting_agent and region_name columns, and append all four parsed results into a single Delta table.

Bronze Layer — spi_bronze_construction_raw

Column	Origin	Description	Storage
contracting_agent	Derived from file	'State & Social Security' or 'Territorial Entities'	String
region_name	Derived from file + row	Nacional, Cataluña, or Madrid	String
period_raw	XLS header	Raw period label	String
indicator_value_raw	XLS cell	Raw numeric value (as text)	String
indicator_category	XLS row label	Construction sub-category	String
_ingestion_timestamp	System	UTC timestamp	Timestamp
_source_file	System	XLS file name	String

Silver Layer — spi_silver_construction_cleaned

Column	Transformation Rule	Description
region_name	Normalized: Nacional, Cataluña, Madrid	Standardized geographic entity
indicator_category	Cleaned from raw label. Subtotal rows removed	Construction sub-category
contracting_agent	Preserved from Bronze	'State & SS' or 'Territorial Entities'
year	Parsed from period_raw, cast to integer	Year
month	Parsed from period_raw, cast to integer	Month (1-12)
indicator_value	Cast to decimal	Tender value (thousands EUR)

Gold mapping: source_code = 'CONSTRUCTION'. The contracting_agent is concatenated into indicator_category for the dim_indicator lookup (e.g., 'Total Construcccion - State & Social Security').

XLSX

Source 5: Tax Revenue by Regional Offices

Agencia Estatal de Administración Tributaria (AEAT)

The tax revenue data arrives as a large multi-sheet XLSX workbook (~6 MB). Only the 'datos_delegaciones' sheet is in scope. The data is in crosstab format where columns represent geographic entities. An unpivot is required to transform the crosstab into flat tabular structure.

Bronze Layer — spi_bronze_tax_raw

The `datos_delegaciones` sheet is extracted as-is using `openpyxl`, preserving the crosstab structure. All fields stored as string. Column names from the sheet header row.

Silver Layer — spi_silver_tax_cleaned

Column	Transformation Rule	Description
region_name	Derived from unpivot. Only Total (Nacional), Madrid, and Cataluña columns retained	Standardized geographic entity
indicator_category	From tax concept row labels	Tax concept (IRPF, IVA, Sociedades)
year	Parsed from period info, cast to integer	Year
month	Parsed from period info, cast to integer	Month (1-12)
indicator_value	Cell value after unpivot, cast to decimal	Tax revenue (thousands EUR)

Unpivot logic: The crosstab has geographic entities as columns. The transformation selects only Total, Madrid, and Cataluña columns, then unpivots them into a `region_name` column. One source row with three value columns becomes three rows.

Gold mapping: Identical pattern. `source_code` = 'TAX'.

4. Star Schema Specification

The Gold layer implements a unified star schema in the Fabric Warehouse. All five sources converge into a single fact table, enabling cross-domain analysis through shared dimensions.

Changes from Phase 2: (1) percentage_change and absolute_change columns replaced by dim_measure dimension (IDX, YoY, MoM, YTD, QTD as separate rows). (2) dim_source promoted to standalone dimension with 5 rows. (3) Composite primary key on the fact table uses all 5 FKs — no surrogate PK needed.

4.1 Fact Table: spi_fact_indicators

One row per indicator × region × period × measure × source combination. The composite primary key is the combination of all five foreign keys.

Column	Role	Description
indicator_key	FK → dim_indicator	Identifies which indicator this row measures
region_key	FK → dim_region	Identifies the geographic entity
calendar_key	FK → dim_calendar	Identifies the time period (year + month)
measure_key	FK → dim_measure	Identifies the calculation type (IDX, YoY, MoM, YTD, QTD)
source_key	FK → dim_source	Identifies the data source
indicator_value	Measure	Numeric value of the observation or calculation

*Composite PK: (indicator_key, region_key, calendar_key, measure_key, source_key). No surrogate PK needed.
Volume estimate: ~360,000 rows.*

4.2 Dimension: spi_dim_indicator

Catalog of all indicators across the five sources. Each row represents a unique indicator.

Column	Description	Example Values
indicator_key	Surrogate key (auto-generated)	1, 2, 3, ...
indicator_name	Full descriptive name	'IPC - Alimentacion y bebidas'
indicator_category	Sub-classification within domain	'Alimentacion y bebidas'
domain	Thematic domain	'Consumer Prices', 'Energy', etc.
unit_of_measure	Unit of the original value	'Index base 2025', 'MWh', 'Thousands EUR'
frequency	Update frequency	'Monthly' (all)

Sample dimension content (representative rows):

key	indicator_name	domain	unit_of_measure
1	IPC - Alimentacion y bebidas	Consumer Prices	Index base 2025
15	IPI - Bienes de consumo	Industrial Production	Index base 2021
25	Energy - Demanda real	Energy	MWh
26	Construction - Total - State & SS	Construction	Thousands EUR
35	Tax - IRPF	Tax Revenue	Thousands EUR

The complete catalog will be finalized during development (Phase 5). Estimated total: ~40-60 rows.

4.3 Dimension: spi_dim_region

Three geographic entities. Fixed, manually-maintained dimension.

region_key	region_code	region_name	region_type
1	NAC	Nacional	Aggregated National
2	CAT	Cataluña	Autonomous Community
3	MAD	Madrid	Autonomous Community

4.4 Dimension: spi_dim_calendar

Time dimension at monthly granularity. Generated programmatically (2000 to current year).

Column	Description	Example
calendar_key	Surrogate key derived from YYYYMM	202603
year	Calendar year	2026
month	Calendar month (1-12)	3
month_name	English month name	'March'
quarter	Calendar quarter (1-4)	1
year_month	Formatted YYYY-MM for display	'2026-03'
is_current_year	Boolean flag for current calendar year	True / False

Generation rule: Every month from January 2000 through December of the current year. Total: ~324 rows.

4.5 Dimension: spi_dim_measure

Five measure types. Fixed, manually-maintained dimension.

key	code	name	description
1	IDX	Index / Original Value	Original value as provided by the source
2	YoY	Year-over-Year Rate	% change vs. same month of previous year
3	MoM	Month-over-Month Rate	% change vs. immediately preceding month
4	YTD	Year-to-Date Cumulative	Cumulative % change from Jan to current month vs. same period of previous year
5	QTD	Quarter-to-Date Cumulative	Cumulative % change from quarter start vs. same period of previous year

4.6 Dimension: spi_dim_source

Five data sources, one row per source. Supports Power BI navigation and provides institutional metadata.

key	code	name	institution	domain
1	IPC	Consumer Price Index	INE	Consumer Prices
2	IPI	Industrial Production Index	INE	Industrial Production
3	ENERGY	Electricity Demand	REE (Redeia)	Energy
4	CONSTRUCTION	Public Construction Tenders	MITMA	Construction
5	TAX	Tax Revenue by Delegations	AEAT	Tax Revenue

5. Derived Measures — Business Rules

Four derived measures are calculated for every indicator-region-period combination in the Gold layer notebook. Each produces additional rows in spi_fact_indicators.

Calculation scope: Derived measures are calculated per indicator_key and region_key. The window operates on a single indicator in a single region. Cross-indicator or cross-region calculations are not performed. All derived measures depend only on IDX rows, never on each other.

5.1 YoY — Year-over-Year Rate

Definition: Percentage change vs. the same month of the previous year.

Formula: $YoY = ((Value[current] - Value[same\ month,\ prev\ year]) / Value[same\ month,\ prev\ year]) \times 100$

Scenario	Behavior
Previous year value exists and is non-zero	Calculate normally
Previous year value does not exist	Result is NULL (row not generated)
Previous year value is zero	Result is NULL (avoid division by zero)
Current month value is NULL	Result is NULL

Example: IPC Alimentacion, Cataluña: Mar 2026 = 112.5, Mar 2025 = 108.3. $YoY = ((112.5 - 108.3) / 108.3) \times 100 = 3.88\%$

5.2 MoM — Month-over-Month Rate

Definition: Percentage change vs. the immediately preceding month.

Formula: $MoM = ((Value[current] - Value[prev\ month]) / Value[prev\ month]) \times 100$

Scenario	Behavior
Previous month exists and is non-zero	Calculate normally
Previous month does not exist (first in series)	Result is NULL
Previous month is zero	Result is NULL
January	Uses December of the previous year

5.3 YTD — Year-to-Date Cumulative Rate

Definition: Cumulative % change from January through current month, vs. same period of previous year.

Formula: $YTD = ((Sum[Jan..current,\ this\ year] - Sum[Jan..current,\ prev\ year]) / Sum[Jan..current,\ prev\ year]) \times 100$

Scenario	Behavior
All months exist for both years	Calculate normally
Any month missing in previous year period	Result is NULL for that month

Scenario	Behavior
January	YTD equals YoY (single month accumulation)
Previous year cumulative sum is zero	Result is NULL

5.4 QTD — Quarter-to-Date Cumulative Rate

Definition: Cumulative % change from quarter start through current month, vs. same period of previous year.

Quarter boundaries: Q1 = Jan-Mar, Q2 = Apr-Jun, Q3 = Jul-Sep, Q4 = Oct-Dec.

Scenario	Behavior
All months in quarter exist for both years	Calculate normally
Any month missing in previous year quarter	Result is NULL
First month of quarter (Jan, Apr, Jul, Oct)	QTD equals YoY for that month
Previous year quarterly sum is zero	Result is NULL

5.5 Calculation Order

All four measures depend only on IDX rows and are independent of each other. The Gold notebook: (1) loads all IDX rows from Silver, (2) computes YoY, MoM, YTD, QTD in parallel, (3) writes all rows (IDX + derived) to spi_fact_indicators in a single operation.

6. Data Quality & Operational Rules

6.1 Data Quality Validations

#	Rule	Layer	Action on Failure
DQ-01	Every fact row must have valid FKs to all five dimensions	Gold	Reject row; log error
DQ-02	No duplicates: composite PK must be unique	Gold	Keep latest; log warning
DQ-03	indicator_value must not be NULL for measure_code = 'IDX'	Gold	Reject row; log error
DQ-04	Derived measures can be NULL only per business rules in Section 5	Gold	Informational only
DQ-05	Every source must produce at least one IDX row per region in most recent period	Gold	Log warning
DQ-06	Silver tables must contain only: Nacional, Cataluña, Madrid	Silver	Reject row; log error
DQ-07	Year between 2000 and current year; month between 1 and 12	Silver	Reject row; log error
DQ-08	No negative values for IPC, IPI, and Energy indicators	Silver	Log warning; preserve row
DQ-09	Construction and Tax: log warning on negative values but preserve	Silver	Log warning; preserve row

6.2 Logging Framework

All pipeline and notebook executions are logged to spi_log_pipeline_execution in the Lakehouse.

Event	Trigger	Status
Pipeline start	Beginning of any pipeline or notebook	running
Pipeline success	Successful completion	success
Pipeline failure	Unhandled error	failed
DQ warning	DQ rule triggered but row preserved	success (warning in error_message)

Column	Description	Example
pipeline_name	Name of the pipeline or notebook	'spi_nb_gold_load'
layer	Medallion layer	'gold'
source	Source identifier or 'all'	'ipc', 'energy', 'all'
status	Execution status	'running', 'success', 'failed'
start_time	Start timestamp (UTC)	'2026-04-10T14:30:00Z'
end_time	End timestamp (UTC, null if running)	'2026-04-10T14:32:15Z'
error_message	Error details or DQ warnings	'DQ-08: 3 negative values in IPC'

Column	Description	Example
rows_processed	Rows written (null if failed)	45230

6.3 Operational Behavior

- **Failure isolation:** Sources are independent. If one fails within a layer, the rest continue. If a source's Bronze fails, its Silver and Gold steps are skipped.
- **Idempotency:** Every layer uses truncate-and-reload (except initial API load). Re-executing produces identical results with no duplicate risk.
- **Retry:** Only the REST API source uses retry (3 attempts, 30s intervals). File and database sources do not retry — their failures are structural.

7. Power BI Report Specification

The Power BI report connects to the Fabric Warehouse via Direct Lake. PBIP format for Git version control. The report has six pages: one Overview and five domain-specific detail pages.

7.1 Report Structure

The report uses a hub-and-spoke navigation pattern: one Overview page as the entry point, and five dedicated detail pages with domain-specific visuals.

Page	Title	Purpose
1	Overview	Executive summary: one YoY card per source for the most recent period
2	Consumer Prices (IPC)	ECOICOP category breakdown, regional inflation comparison
3	Industrial Production (IPI)	Economic destination breakdown, industrial cycle by region
4	Energy Demand	Demand evolution, regional comparison, seasonal patterns
5	Construction Tenders	Contracting agent split (State vs. Territorial), regional comparison
6	Tax Revenue	Tax concept hierarchy breakdown, regional comparison

7.2 Page 1: Overview

Visual	Content	Interaction
Title bar	'Spain Economic Indicators — Overview'	None
5 × KPI Cards	Source name, latest YoY rate (Nacional), period label, trend arrow	Click = drill through to detail page
Period slicer	Year-month selector	Updates all cards
Region slicer	Dropdown: Nacional, Cataluña, Madrid	Updates all cards

7.3 Shared Visual Template

All five detail pages share a base visual layout:

Visual	Content	Notes
Page title	Source name and domain	Fixed per page
KPI Cards (3-4)	Latest IDX, YoY, MoM, YTD	Reference period from slicer
Line chart	Time series of IDX, one line per region	Cross-highlight with category slicer
Bar chart	YoY comparison across categories	Sorted descending by absolute value
Region slicer	Dropdown: Nacional, Cataluña, Madrid	Filters all visuals
Measure slicer	Toggle: IDX, YoY, MoM, YTD, QTD	Changes measure in charts
Back button	Navigation to Overview	Top-left corner

7.4 Domain-Specific Visuals

Page	Domain-Specific Visual	Rationale
IPC	Stacked bar: ECOICOP weight contribution	Shows which categories drive inflation
IPC	Heatmap: category × month with YoY color	Reveals seasonal inflation patterns
IPI	Multi-line chart: one line per econ. destination	Compares industrial sectors within region
IPI	Clustered bar: regional comparison per destination	Cataluña vs. Madrid structure
Energy	Area chart: demand evolution with YoY overlay	Shows trend and growth simultaneously
Energy	Column chart: monthly seasonal pattern	Highlights summer/winter demand peaks
Construction	Stacked bar: State vs. Territorial per region	Shows contracting agent split
Construction	Waterfall chart: period-over-period change	Where public investment grows/declines
Tax	Treemap: tax concept hierarchy by volume	Relative weight of IRPF, IVA, etc.
Tax	Matrix: concepts × periods with cond. formatting	Tabular view with color-coded growth

7.5 Key DAX Measures

Measure Name	Business Logic
Latest Value	Returns indicator_value for the most recent period with data, filtered by context
Latest Period Label	Returns the year_month label of the most recent period with data
Trend Direction	'Up' if latest > previous, 'Down' if less, 'Flat' if equal. For trend arrows
Selected Measure Value	Returns indicator_value filtered by measure selected in slicer

8. Traceability Matrix

This matrix links Phase 1 requirements to Phase 3 specifications, ensuring complete coverage.

Phase 1 Requirement	Phase 3 Section	Status
Source 1: IPC (CSV) ingestion and transformation	3.1 Source-to-Target: IPC	Specified
Source 2: IPI (Azure SQL) ingestion and transformation	3.2 Source-to-Target: IPI	Specified
Source 3: Energy Demand (REST API) ingestion and transformation	3.3 Source-to-Target: Energy	Specified
Source 4: Construction Tenders (XLS) ingestion and transformation	3.4 Source-to-Target: Construction	Specified
Source 5: Tax Revenue (XLSX) ingestion and transformation	3.5 Source-to-Target: Tax	Specified
Star schema with fact and dimension tables	4.1-4.6 Star Schema Specification	Specified
Unified fact table for cross-domain analysis	4.1 spi_fact_indicators	Specified
Geographic scope: Nacional, Cataluña, Madrid	4.3 spi_dim_region	Specified
Derived measures (percentage change)	5.1-5.5 Derived Measures	Specified
Data quality rules	6.1 Data Quality Validations	Specified
Pipeline orchestration and error handling	6.2-6.3 Logging & Operational Rules	Specified
Power BI report with Direct Lake connectivity	7.1-7.5 Power BI Report Specification	Specified

9. Next Steps

With the functional specification formalized, the project proceeds to **Phase 4: Technical Specification**, which will deliver DDL statements, PySpark code specifications, connection configurations, Dataflow Gen2 step definitions, pipeline orchestration detail, and DAX measure implementations.

Phase Progress

0	Context & Strategy Deliverable: Project Brief (PDF)	COMPLETE
1	Requirements & Source Catalog Deliverable: Requirements Document + Source Catalog	COMPLETE
2	Architecture Design Deliverable: Architecture Design Document	COMPLETE
3	Functional Specification Deliverable: Functional Specification (PDF)	COMPLETE
4	Technical Specification Deliverable: Technical Document (PDF)	NEXT
5	Development in Fabric Deliverable: Working Solution + Code	PENDING
6	GitHub Publication Deliverable: GitHub Repository + README	PENDING

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